CS555 HW1 – Question 1: Observations

1a. Crank–Nicolson with $u^0 = \sin(\pi x)$

Table 1: Euler Backward (EB) — $u^0 = \sin(\pi x)$

N	nstep	L_2 error	ratio
1500	25	7.69e-02	—
1500	50	3.87e-02	1.99
1500	100	1.94e-02	1.99
1500	200	9.72e-03	2.00
1500	400	4.87e-03	2.00
1500	800	2.43e-03	2.00
1500	1600	1.22e-03	2.00

Table 2: Crank–Nicolson (CN) — $u^0 = \sin(\pi x)$

N	nstep	L_2 error	ratio
1500	25	1.03e-03	—
1500	50	2.56e-04	4.00
1500	100	6.40e-05	4.00
1500	200	1.59e-05	4.00
1500	400	3.93e-06	4.06
1500	800	9.34e-07	4.20
1500	1600	2.42e-07	3.86

Observations:

- **EB** shows error ratios converging to **2** as Δt is halved, confirming **1st-order** accuracy in time (error $\propto \Delta t$).
- **CN** shows error ratios converging to **4** as Δt is halved, confirming **2nd-order** accuracy in time (error $\propto \Delta t^2$). Halving Δt reduces the error by a factor of $4 = 2^2$.
- CN is significantly more accurate than EB at the same Δt. For example, at nstep = 100, the CN error is roughly 300× smaller than the EB error.

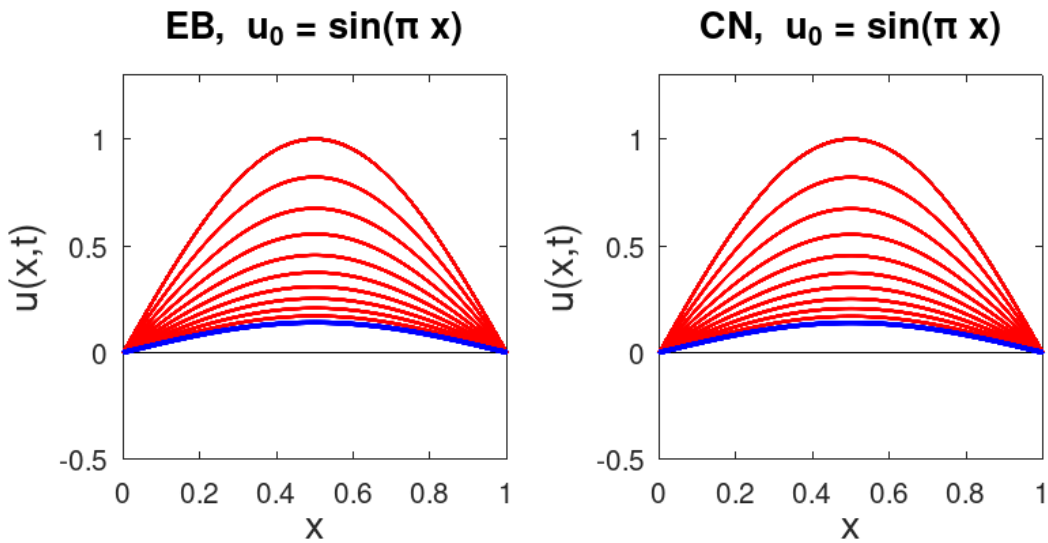

Figure 1: EB (left) vs. CN (right) with $u^0 = \sin(\pi x)$. Both methods produce smooth, oscillation-free solutions for this smooth IC.

1b. Square wave IC: $u^0 = \pi/4$

Table 3: Euler Backward (EB) — $u^0 = \pi/4$

N	nstep	L_2 error	ratio
1500	25	7.69e-02	—
1500	50	3.87e-02	1.99
1500	100	1.94e-02	1.99
1500	200	9.72e-03	2.00
1500	400	4.87e-03	2.00
1500	800	2.43e-03	2.00
1500	1600	1.22e-03	2.00

Observations:

- **EB** still converges with ratio ≈ 2 (1st-order in time) even with the discontinuous IC. EB is **L-stable**: its growth factor $G = 1/(1 + \lambda\Delta t) \to 0$ as $\lambda\Delta t \to -\infty$, so it strongly damps all high-frequency Fourier modes excited by the boundary discontinuity.
- **CN fails to converge** properly. The errors are $O(1)$ and the ratio is ≈ 1.2 instead of 4.

Table 4: Crank–Nicolson (CN) — $u^0 = \pi/4$

N	nstep	L_2 error	ratio
1500	25	2.49e+00	—
1500	50	2.09e+00	1.19
1500	100	1.75e+00	1.19
1500	200	1.47e+00	1.19
1500	400	1.23e+00	1.20
1500	800	1.02e+00	1.20
1500	1600	8.49e-01	1.20

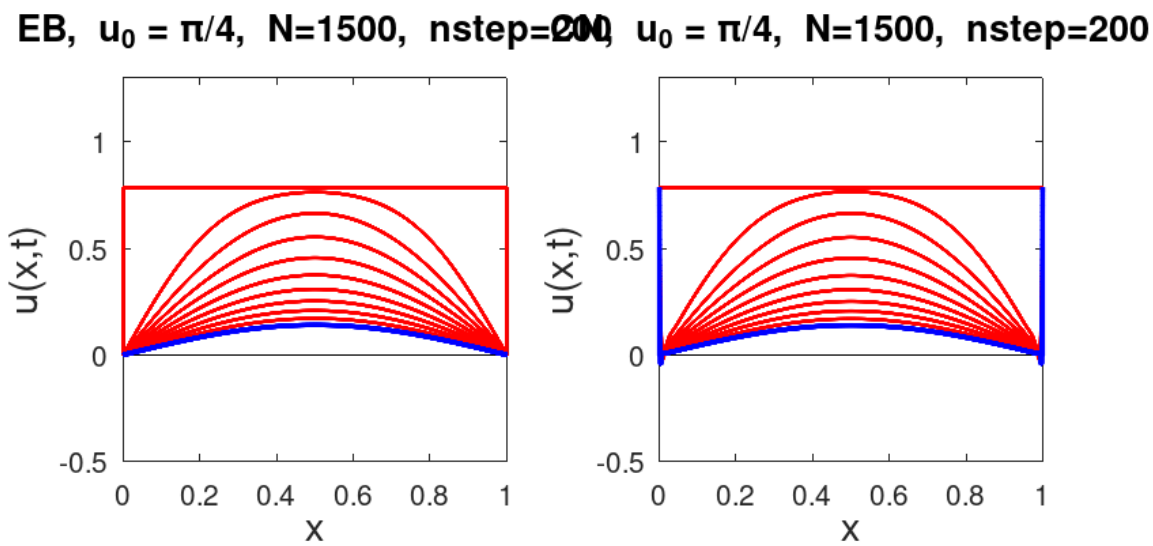

Figure 2: EB (left) vs. CN (right) with $u^0 = \pi/4$. CN exhibits persistent oscillations near the boundaries.

CN is **A-stable but not L-stable**: its growth factor $G = (1 - \lambda\Delta t/2)/(1 + \lambda\Delta t/2) \to -1$ as $\lambda\Delta t \to -\infty$. The high-frequency modes excited by the boundary discontinuity oscillate with amplitude ≈ 1 and are never damped, producing persistent oscillations visible in the plot near $x = 0$ and $x = 1$.

- **Key lesson:** CN's 2nd-order accuracy advantage only applies to **smooth** initial conditions. For non-smooth data (like this square wave), EB's L-stability makes it more robust.

1c. BDF2 (both ICs)

Table 5: BDF2 — $u^0 = \sin(\pi x)$

N	nstep	L_2 error	ratio
1500	25	6.00e-04	—
1500	50	1.45e-04	4.13
1500	100	3.61e-05	4.03
1500	200	9.06e-06	3.98
1500	400	2.33e-06	3.88
1500	800	6.68e-07	3.49
1500	1600	2.85e-07	2.34

Table 6: BDF2 — $u^0 = \pi/4$

N	nstep	L_2 error	ratio
1500	25	6.00e-04	—
1500	50	1.45e-04	4.13
1500	100	3.60e-05	4.03
1500	200	9.00e-06	4.00
1500	400	2.27e-06	3.96
1500	800	6.11e-07	3.72
1500	1600	2.41e-07	2.53

Observations:

- **2nd-order convergence:** BDF2 shows error ratios approaching **4** as Δt is halved for both ICs, confirming 2nd-order temporal accuracy—same order as CN.
- **L-stability advantage:** Unlike CN, BDF2 handles the discontinuous IC ($u^0 = \pi/4$) **without oscillations**. BDF2 is L-stable: $G \to 0$ as $\lambda\Delta t \to -\infty$, strongly damping high-frequency modes just like EB.
- **Both ICs give nearly identical errors:** Since BDF2 (like EB) damps the high-frequency modes that distinguish the two ICs, by $t = 1$ both solutions are dominated by the $k = 1$ Fourier mode and the errors are essentially the same.
- **Ratio degradation at fine Δt:** The ratio drops below 4 at the finest Δt (nstep $= 800, 1600$). This is because the temporal error becomes comparable to the spatial discretization error at $N = 1500$, so further Δt refinement no longer gives pure temporal convergence.
- **BDF2 = best of both worlds:** It combines CN's 2nd-order accuracy with EB's L-stability, making it robust for non-smooth data while maintaining high-order convergence for smooth data.

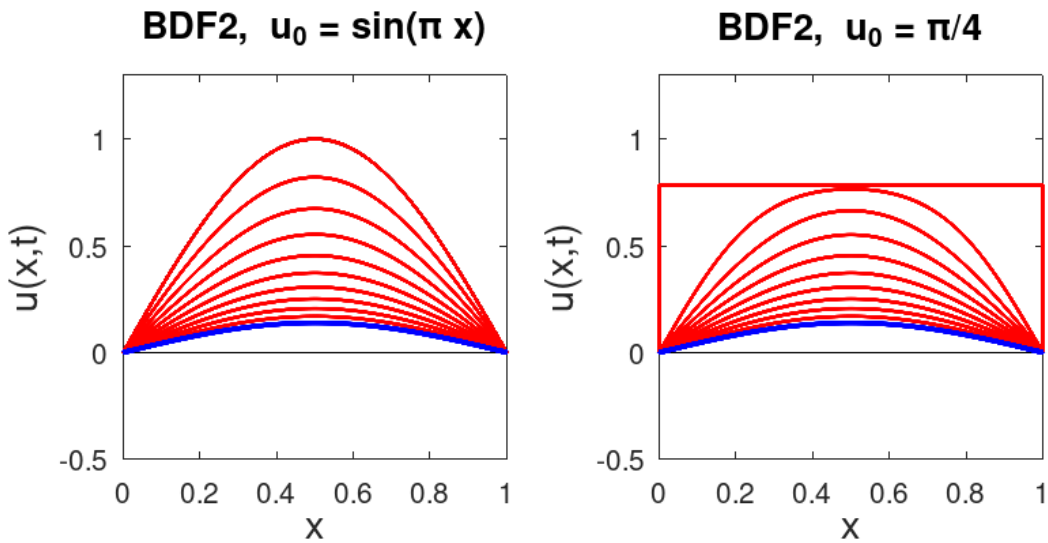

Figure 3: BDF2 with $u^0 = \sin(\pi x)$ (left) and $u^0 = \pi/4$ (right). No oscillations in either case.

Extra Credit: Next leading-order term in the square wave

The square wave $u^0 = \pi/4$ has the Fourier sine series

$$u^0 = \sum_{k=1,3,5,\ldots} \frac{1}{k} \sin(k\pi x),$$

so the exact solution is

$$u(x,t) = \sum_{k=1,3,5,\ldots} \frac{1}{k} e^{-\nu(k\pi)^2 t} \sin(k\pi x).$$

At $t = 1$ with $\nu = 0.2$, the magnitudes of the leading Fourier coefficients are:

k	$\frac{1}{k} e^{-\nu(k\pi)^2}$
1	1.39×10^{-1}
3	$\mathbf{6.42 \times 10^{-9}}$
5	7.40×10^{-23}
7	1.41×10^{-43}

The next leading-order term ($k = 3$) has magnitude $\approx 6.4 \times 10^{-9}$, which is a factor of $\approx 4.6 \times 10^{-8}$ smaller than the dominant $k = 1$ mode. The $k = 3$ mode decays as $e^{-9\nu\pi^2}$ versus $e^{-\nu\pi^2}$ for $k = 1$; the $9\times$ larger exponent causes exponential suppression.

Comparison to FD errors:

- EB errors ($\sim 10^{-3}$): $\sim 10^5$ times **larger** than the $k = 3$ term.
- BDF2 errors ($\sim 10^{-7}$): $\sim 10^2$ times **larger** than the $k = 3$ term.
- CN errors ($\sim 10^0$): completely dominated by oscillations, irrelevant.

Conclusion: The $k = 3$ Fourier truncation error (6.4×10^{-9}) is negligible compared to all FD temporal errors in the preceding questions. This confirms that the assumption $u(x, 1) \approx e^{-\nu\pi^2} \sin(\pi x)$ (the $k = 1$ eigenmode alone) is excellent. The diffusion operator exponentially annihilates higher modes, and the time-stepping error dominates in every case tested.